

4	(a)	(i)	The increase in internal energy of a system is equal to the sum of heat (or thermal energy) supplied to the system (or heating) and the work done on the system.	
		(ii)	Increase in volume (or intermolecular separation) is more significant in vaporisation compared to melting Hence, the increase in internal energy and work done by the substance in expanding against atmospheric pressure is greater in vaporisation compared to melting resulting in a larger specific latent heat of vaporisation compared to specific latent heat of fusion for the same substance.	
	(b)	(i)	At point A: $p_A V_A = 1.0 \times 10^5 \times 4.5 \times 10^{-2} = 4500$ At point B: $p_B V_B = 3.0 \times 10^5 \times 1.5 \times 10^{-2} = 4500$ Since (from the equation of state of an ideal gas) pV is directly proportional to the thermodynamic temperature of the gas, it follows that $p_A V_A = p_B V_B$ means the temperature at A and B are the same and hence A $\rightarrow$ B is an isothermal process	

		(ii)	pressure = $\frac{\text{Force}}{\text{Area}}$ and Force = $\Delta v \frac{Nm}{t} = (m\Delta v) \left(\frac{N}{t}\right) = (\Delta p) \left(\frac{N}{t}\right)$ From A→B, the temperature is constant, increase in pressure is due to the <u>increase in the rate of collision between the gas molecules and the walls of the container due to the decrease in the volume of the gas.</u> From C→B, the volume of the gas is constant while temperature of the gas is increased resulting in the increase in mean square speed of the gas molecules. This <u>increases the rate of collision</u> as well as <u>the change in momentum of each molecule as it collides with the wall.</u>	
		(iii)	Area under the graph $\approx \frac{1}{2} (1 + 2.25) \times 10^5 \times (4.5 - 1.5) \times 10^{-2} = 4900 \text{ J}$ Since $q + w = 0$ $q = -w = -4900 \text{ J}$ Heat removed = 4900 J (accept 4000 to 5500 J)	